Name ______________________________ Section ________

MTH:2310 DISCRETE MATH
QUIZ 4
DUE: MONDAY MAR. 3

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(9 points) Sets*
Describe each of the following sets using roster notation.

(a) *(3 points)* $A = \{x | (x \text{ is a U.S. state}) \wedge (x \text{ is in the Midwest})\}$

{ND, SD, NE, KS, MN, IA, MO, WI, IL, MI, IN, OH}
According to US Census 2020

(b) *(3 points)* $B = \{x | (x \in \mathbb{N}) \wedge (x \text{ is less than } 20) \wedge (x \text{ is divisible by } 3)\}$

$\{0, 3, 6, 9, 12, 15, 18\}$

(c) *(3 points)* $C = \{x | (x = 2n + 1) \wedge (n \in \mathbb{N})\}$

$\{1, 3, 5, 7, 9, \dots\}$

Problem 2. *(6 points) Sets*
Determine the cardinality of the following sets.

(a) *(2 points)* $A = \{x | (x \text{ is a U.S. state}) \wedge (x \text{ is in the Midwest})\}$

$|A| = 12$

(b) *(2 points)* $B = \{x | (x \in \mathbb{N}) \wedge (x \text{ is less than } 20) \wedge (x \text{ is divisible by } 3)\}$

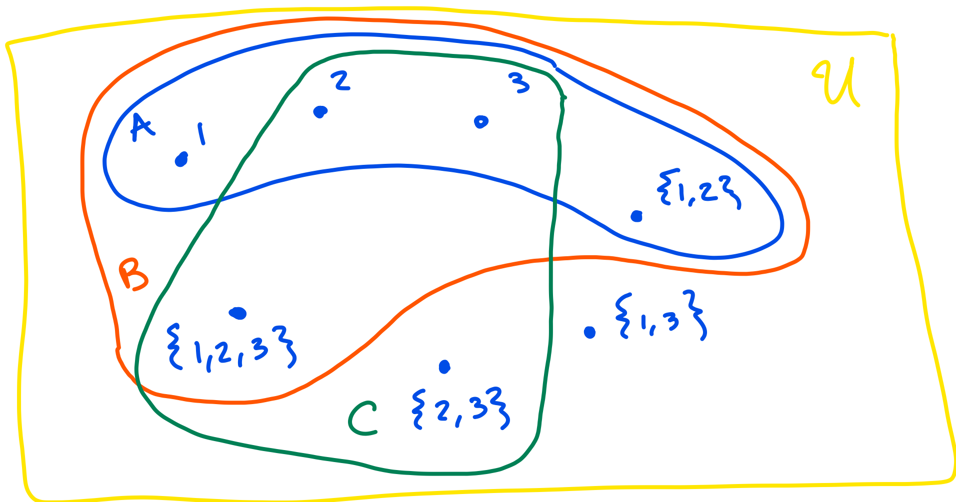
$|B| = 7$

(c) *(2 points)* $C = \{x | (x = 2n + 1) \wedge (n \in \mathbb{N})\}$

$|C| = \infty$

Problem 3. *(10 points) Sets*
Draw a Venn diagram of the following sets U, A, B, C. Make sure your diagram is fully labeled.

$$U = \{1, 2, 3, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$
$$A = \{1, 2, 3, \{1, 2\}\}$$
$$B = \{\{1, 2\}, \{1, 2, 3\}, 3, 1, 2\}$$
$$C = \{\{1, 2, 3\}, 2, 3, \{2, 3\}\}$$



Name ____ Section ________

Problem 4. *(16 points) Sets*
List all subsets of the following sets.

(a) *(8 points)* $A = \{1, 2, 3, \{1, 2\}\}$

$\emptyset$

$\{1\}$ $\{2\}$ $\{3\}$ $\{1,2\}$

$\{1,2\}$ $\{1,3\}$ $\{1,\{1,2\}\}$

$\{2,3\}$ $\{2,\{1,2\}\}$ $\{3,\{1,2\}\}$

$\{1,2,3\}$ $\{1,2,\{1,2\}\}$ $\{1,3,\{1,2\}\}$ $\{2,3,\{1,2\}\}$

$\{1,2,3,\{1,2\}\}$

(b) *(8 points)* $B = \{\{1, 2, 3\}, 2, 3, \{2, 3\}\}$

$\emptyset$

$\{\{1,2,3\}\}$ $\{2\}$ $\{3\}$ $\{\{2,3\}\}$

$\{\{1,2,3\},2\}$ $\{\{1,2,3\},3\}$ $\{\{1,2,3\},\{2,3\}\}$

$\{2,3\}$ $\{2,\{2,3\}\}$ $\{3,\{2,3\}\}$

$\{\{1,2,3\},2,3\}$ $\{\{1,2,3\},2,\{2,3\}\}$ $\{\{1,2,3\},3,\{2,3\}\}$ $\{2,3,\{2,3\}\}$

$\{\{1,2,3\},2,3,\{2,3\}\}$

Problem 5. *(8 points) Sets*
Determine whether these statements are true or false.

(a) *(2 points)* $\emptyset \in \{\emptyset\}$

T

(b) *(2 points)* $\emptyset \subseteq \{\emptyset\}$

T

$\emptyset$ is a subset of every set

(c) *(2 points)* $\{\emptyset\} \in \{\emptyset, \{\emptyset, \emptyset\}, \emptyset\}$

T

$$\{\emptyset, \{\emptyset, \emptyset\}, \emptyset\}$$
$$\boxed{= \{\emptyset, \{\emptyset\}\}}$$

(d) *(2 points)* $\{\{\emptyset\}, \emptyset\} \subseteq \{\emptyset, \{\emptyset, \emptyset\}, \emptyset\}$

T

same reason.